

The Fire Kit

What to pack so you can make warmth from nothing

The reel showed you the spark. Here is the whole system — what to carry, how to prepare it, and how to use it in the rain when nothing else works.

CHOOSING YOUR FIRE STEEL

- **Rod length** — 4-inch minimum for field use. Longer rods have more surface area and last far more strikes (a 4-inch rod = ~12,000 strikes).
 - **Ferrocium vs. magnesium** — ferrocium ("ferro rod") produces sparks at ~3,000 °C. Magnesium blocks shave off powder for a longer burn but need more setup. Start with ferrocium.
 - **Striker blade** — the spine of a fixed-blade knife, a dedicated scraper, or any high-carbon steel edge. Stainless steel works poorly. Carbon steel bites the rod and throws sparks.
 - **Practice dry** — strike toward the tinder, not away from it. Short, controlled scrapes beat long wild swipes.
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THE PETROLEUM-JELLY COTTON BALL — STEP BY STEP

- Take a standard cotton ball and pull it slightly open to expose the fibers inside.
 - Scoop a pea-sized amount of petroleum jelly (Vaseline) onto your fingertip.
 - **Work the jelly into the interior fibers** — not just the surface. The jelly coats the strands and becomes the fuel; the cotton is only the wick.
 - **Store pre-made** — keep 10–15 in a small zip-lock bag or a 35mm film canister in your kit. They are ready to use immediately without unwrapping.
 - **In the field** — pull the ball slightly apart to expose uncoated fibers at the center. These dry fibers catch the spark first, then the jelly catches and sustains the flame.
 - **Burn time** — a well-prepared jelly cotton ball burns for 4–8 minutes. Enough to catch damp kindling.
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WHAT TINDER TO LOOK FOR OUTDOORS

- **Dry grass** — fine, dry grass or plant fibers. Roll into a loose bird's-nest bundle.
 - **Birch bark** — contains oils that burn even when damp on the outside. Peel the thin papery inner layers.
 - **Dead pine needles** — gather from beneath the tree, not off the ground where they absorb moisture.
 - **Cattail fluff** — extremely fine and catches sparks almost instantly. Dry season only.
 - **Char cloth** — cotton fabric carbonized slowly in a sealed tin. Catches a spark at near-zero effort and is the single best improvised tinder. Carry 6–8 squares.
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BUILDING FIRE IN WET CONDITIONS

- **Find shelter from wind first** — wind steals heat before the kindling catches. A large tree, rock face, or your own body can block it.
 - **Prepare your tinder platform** — a flat piece of dry bark off the ground keeps tinder away from wet ground.
 - **Layer your fuel in advance** — have pencil-thin kindling and finger-thick sticks ready before you light anything. Never wait until you have flame.
 - **Light from inside out** — place the lit jelly cotton ball at the center of your tinder bundle. Build kindling over it like a tent so heat rises into the fuel.
 - **Wet wood will burn** — split logs down the center to expose dry interior wood. Use the dry core for kindling; the outer shell can fuel the fire once it is established.
 - **Feed slowly** — add each piece one at a time. A rushed fire collapses; a patient one builds.
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YOUR GO-BAG FIRE KIT (WHAT TO PACK)

- **1 ferrocerium rod** — 4-inch minimum, with striker.
 - **10 petroleum-jelly cotton balls** — in a zip-lock bag.
 - **6 squares of char cloth** — in a small metal tin.
 - **1 fixed-blade knife** — carbon steel spine for striking.
 - **A handful of dry tinder** — birch bark strips or fine dry grass in a sealed bag.
 - **1 waterproof match container** — backup only. Matches fail in rain; the ferro rod does not.
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EXTINGUISHING SAFELY

- **Never walk away from a fire that is not fully out** — embers can survive hours and reignite in wind.
 - **Water method** — pour water slowly onto coals, stir with a stick until no steam rises. Feel with the back of your hand 6 inches above — if you feel heat, add more water.
 - **Dirt method** — if no water is available, smother with dry mineral soil (not leaf litter, which can smolder). Stir and repeat until the ground is cool to the touch.
 - **Pack it out** — scatter cold ash; leave no trace. A fire ring should be cold enough to palm before you leave.
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KEEP THIS IN MIND

One jelly cotton ball lit under wet kindling will outlast ten attempts with a lighter.
Prepare 10 before your next trip. They weigh almost nothing.

If this was useful, comment LAMP next Sunday — there's more where this came from.

SOURCES U.S. Forest Service — Leave No Trace fire guidelines · NOLS (National Outdoor Leadership School) — Wilderness Skills Manual · Ferrocerium rod specifications: industry standard test data (12,000+ strikes per 4-inch rod)

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